Session 3.4: Data Integration III – Protein structural analysis

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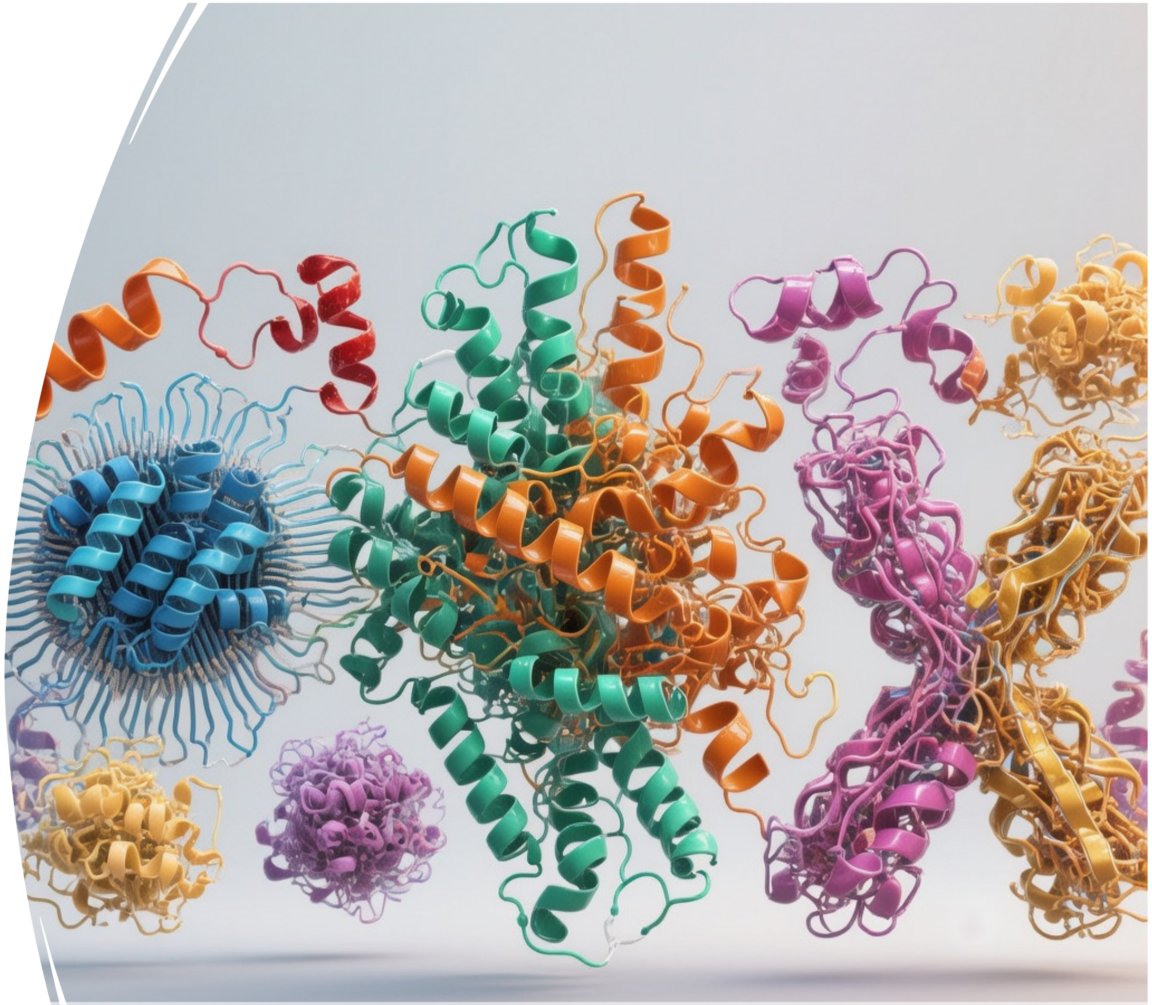
University of Edinburgh

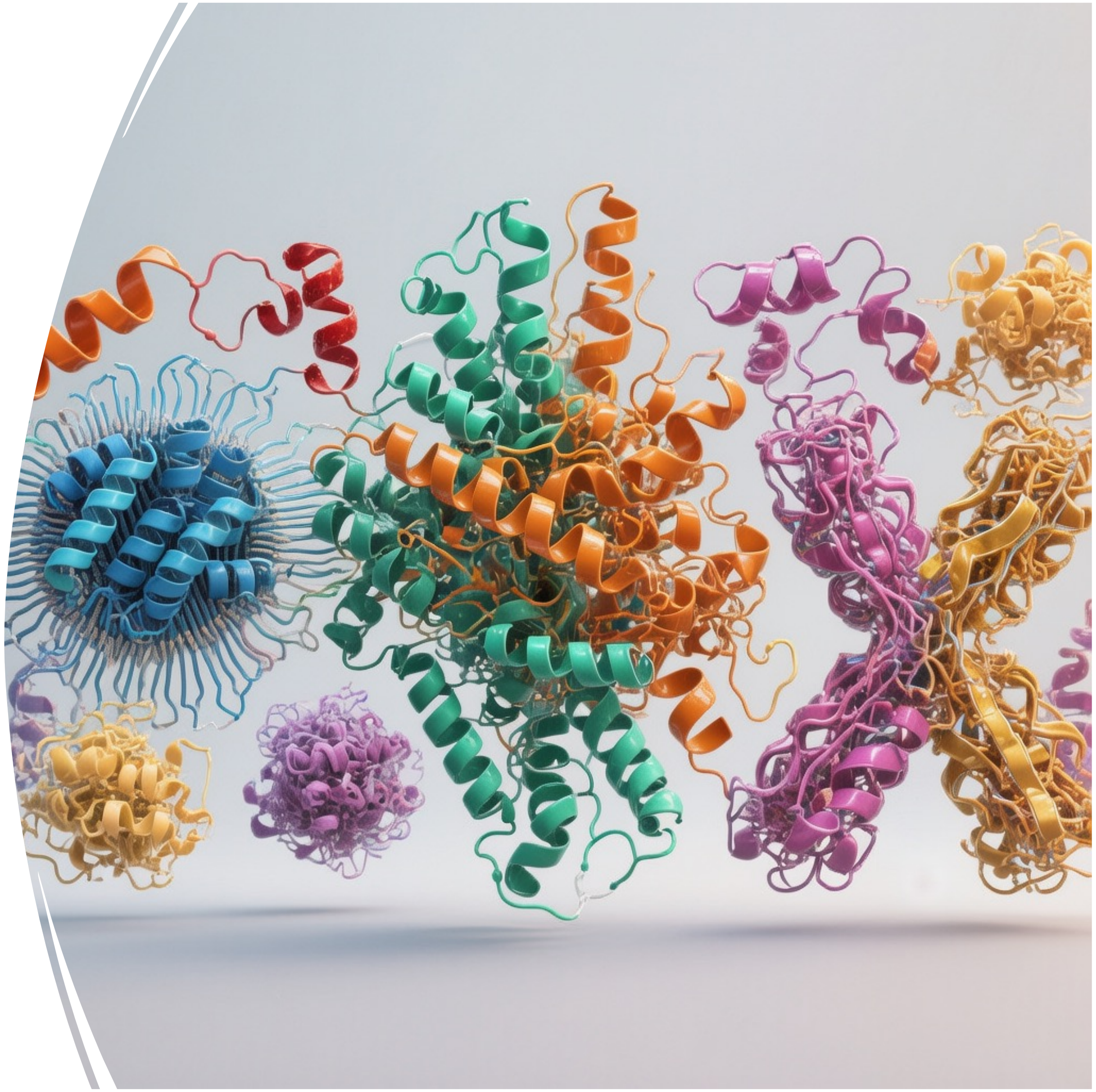


Links and datasets

https://sites.google.com/view/mave2025-lectures

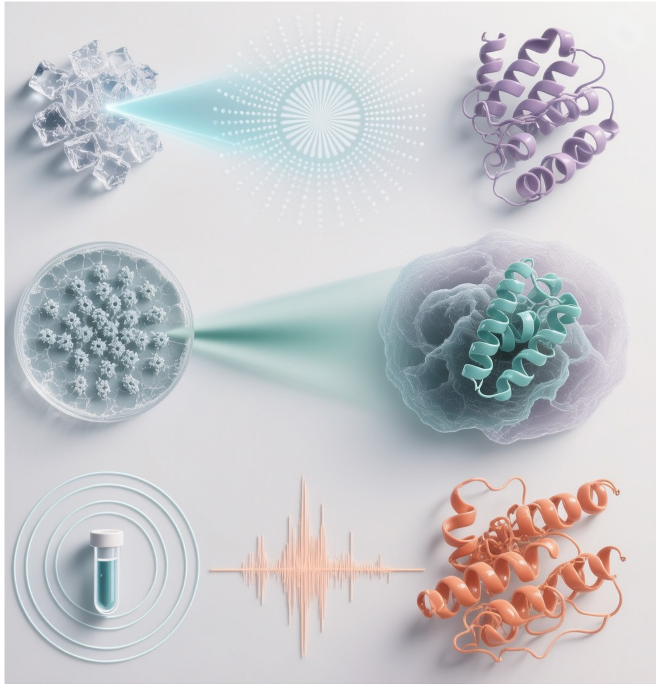
Why care about protein structure?

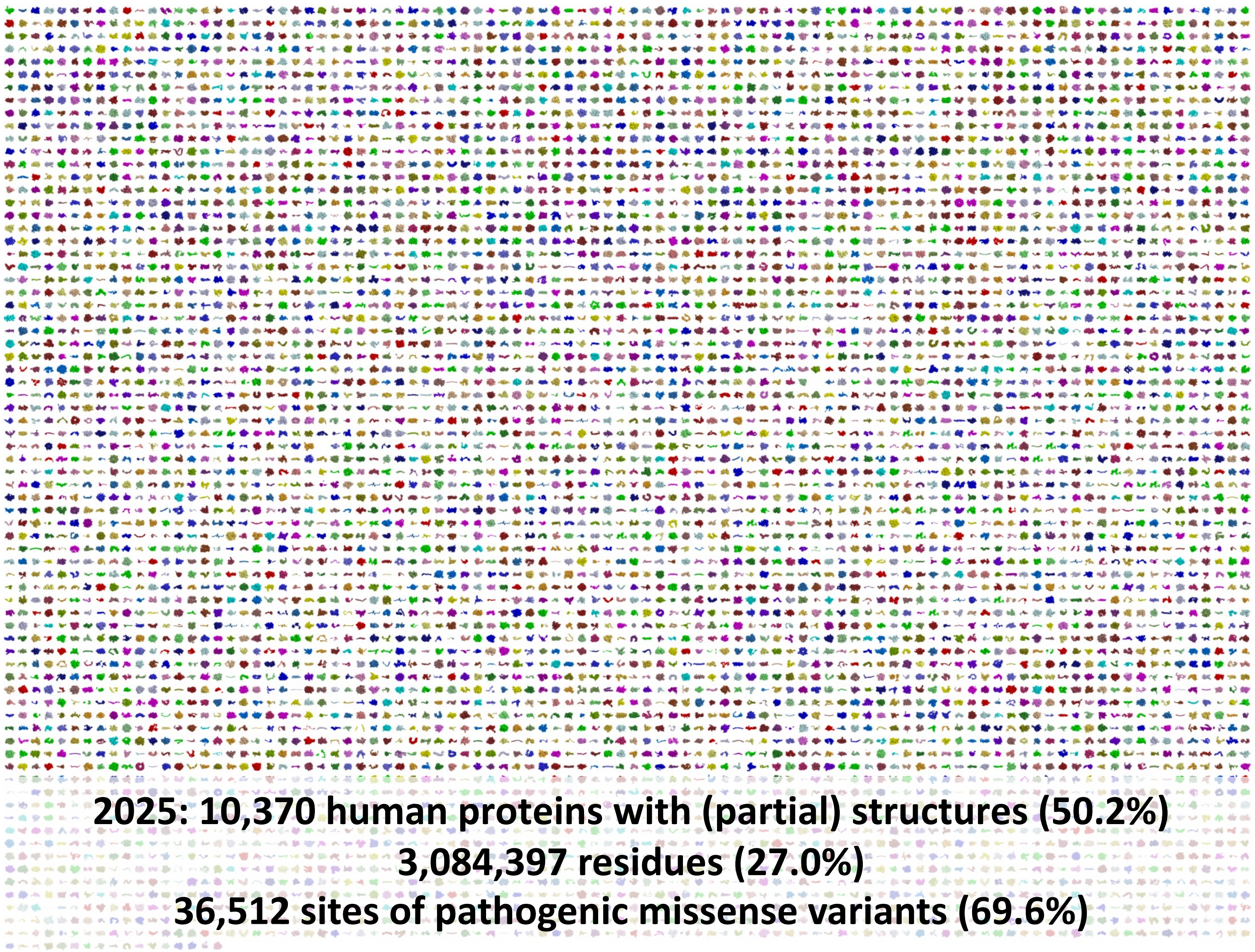
- Most current MAVEs probe proteins, and most pathogenic variants affect coding regions
- The shape of a protein determines what it does
- Structure reveals how variants disrupt function: stability, binding, catalysis, regulation
- Sequence tells you the parts; structure shows their spatial relationships
- Many MAVE patterns only make sense in 3D context
- Visualising structure helps generate mechanistic and therapeutic hypotheses



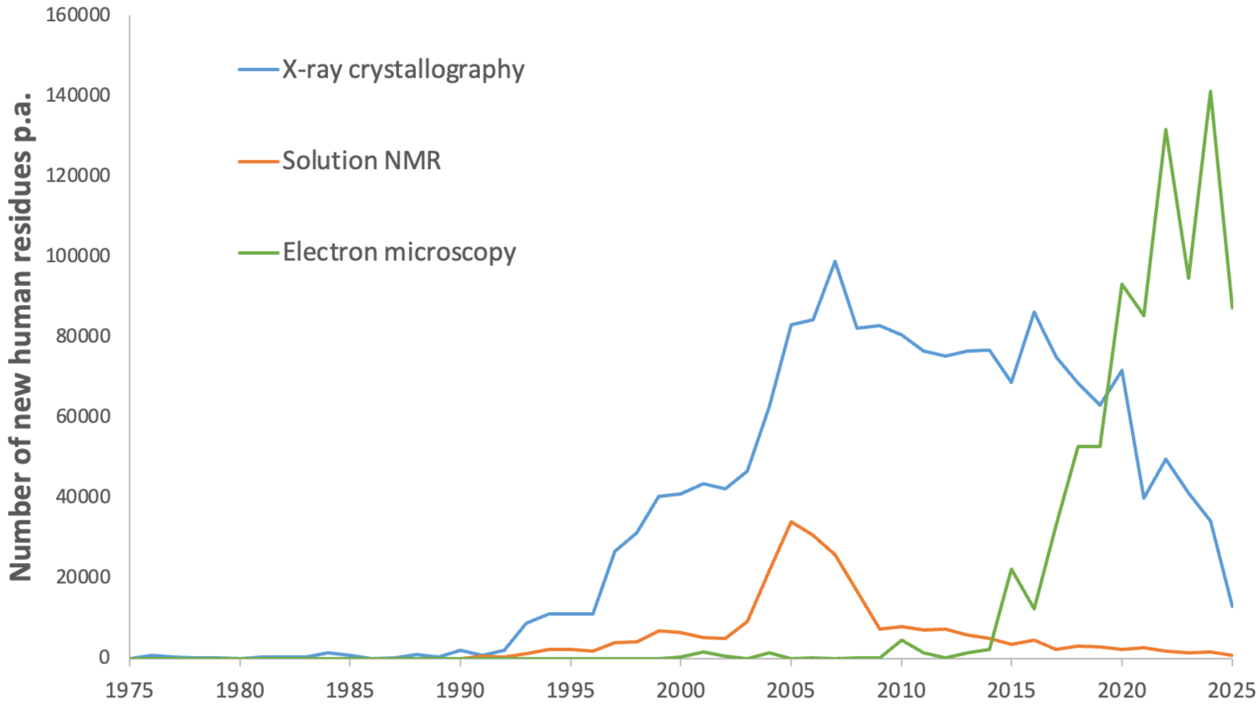
Experimentally determined protein structures

- Structures are models built from experimental measurements of atomic positions.
- **X-ray crystallography** – still most common; high resolution but requires crystals and may miss flexible regions.
- **Cryo-electron microscopy (cryoEM)** – excellent for large complexes; improving rapidly in resolution.
- **NMR spectroscopy** – captures solution dynamics; works best for small, flexible proteins.




2025: 10,370 human proteins with (partial) structures (50.2%)
3,084,397 residues (27.0%)
36,512 sites of pathogenic missense variants (69.6%)

Increasing structural coverage of the human proteome



The Protein Data Bank

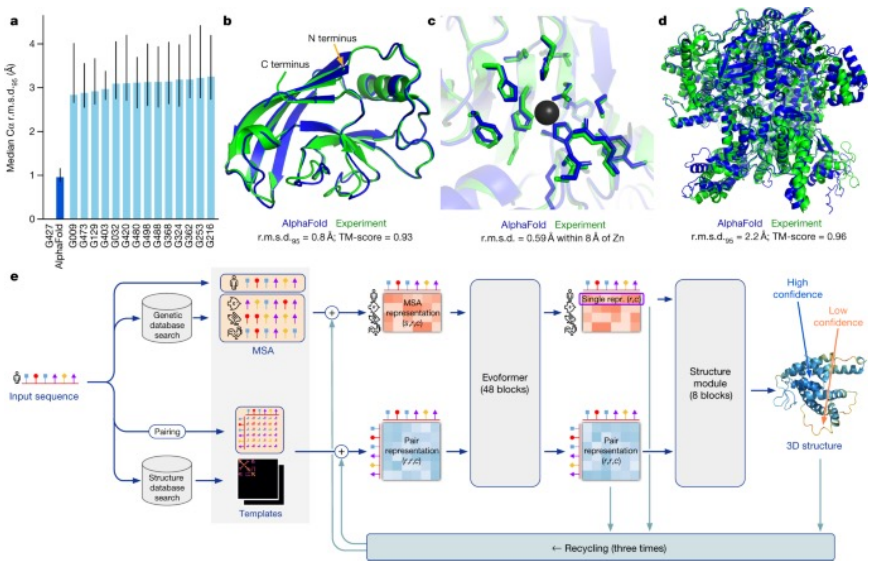
- Global archive of experimentally determined protein, nucleic acid, and complex structures.
- Established in 1971; now contains over 200,000 entries.
- Fully open and standardised — anyone can download coordinates, metadata, and validation data.
- A model for community sharing: similar principles underpin MaveDB.
- Open data enabled the training of AlphaFold and other structure-prediction methods.

Article | Open access | Published: 15 July 2021

Highly accurate protein structure prediction with AlphaFold

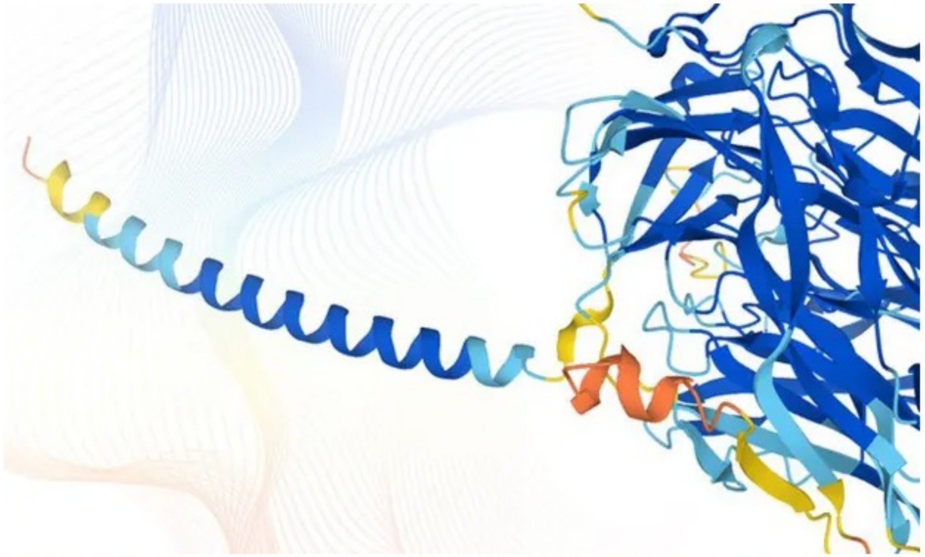
John Jumper, Richard Evans, Alexander Pritzel, Tim Green, Michael Figurnov, Olaf Ronneberger, Kathryn Tunyasuvunakool, Russ Bates, Augustin Žídek, Anna Potapenko, Alex Bridgland, Clemens Meyer, Simon A. A. Kohl, Andrew J. Ballard, Andrew Cowie, Bernardino Romera-Paredes, Stanislav Nikolov, Rishub Jain, Jonas Adler, Trevor Back, Stig Petersen, David Reiman, Ellen Clancy, Michal Zielinski, ... Demis Hassabis + Show authors

Nature **596**, 583–589 (2021) | Cite this article



AI firm DeepMind puts database of the building blocks of life online

AlphaFold program's prediction of nearly 20,000 human protein structures now free for researchers



▲ The AlphaFold database will increase our understanding of how proteins function, say scientists. Photograph: Karen Arnott/EMBL-EBI/PA

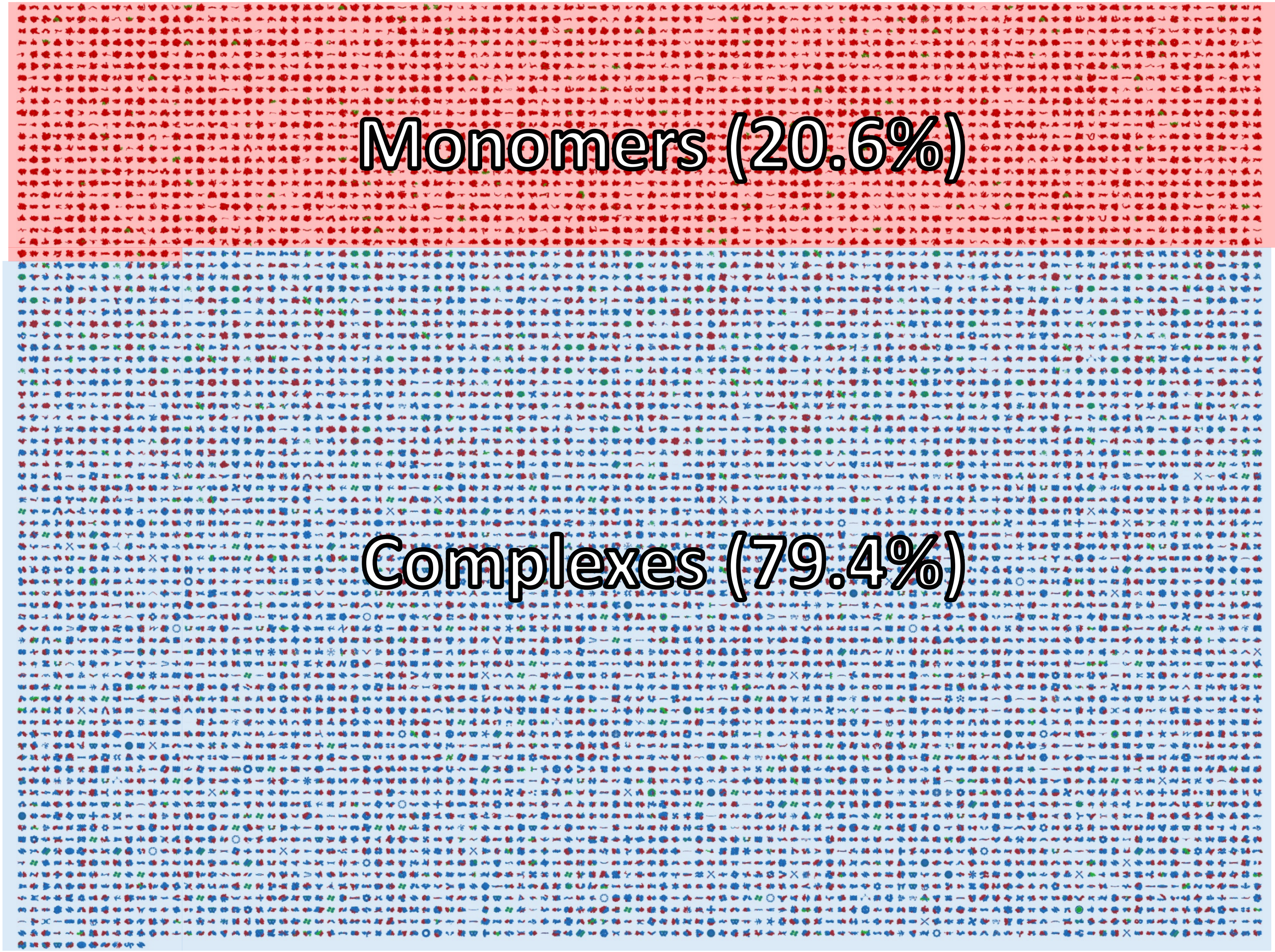
AlphaFold: strengths and limitations

Strengths

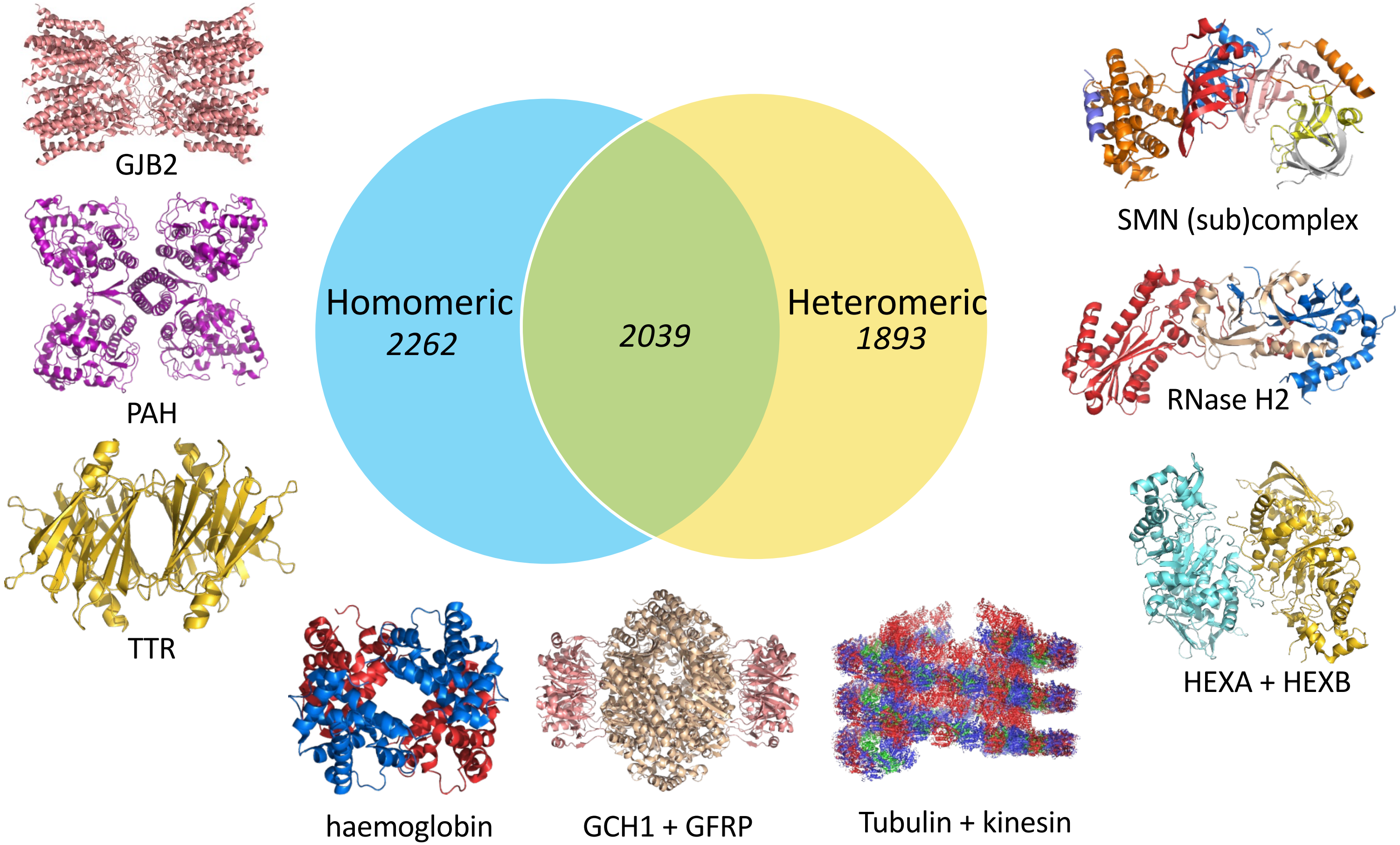
- Provides structures for nearly all known proteins, including those never solved experimentally.
- Often accurate for well-ordered domains.
- Full-length coverage allows viewing domain organisation and interfaces.
- Confidence metrics (pLDDT, PAE) guide interpretation.

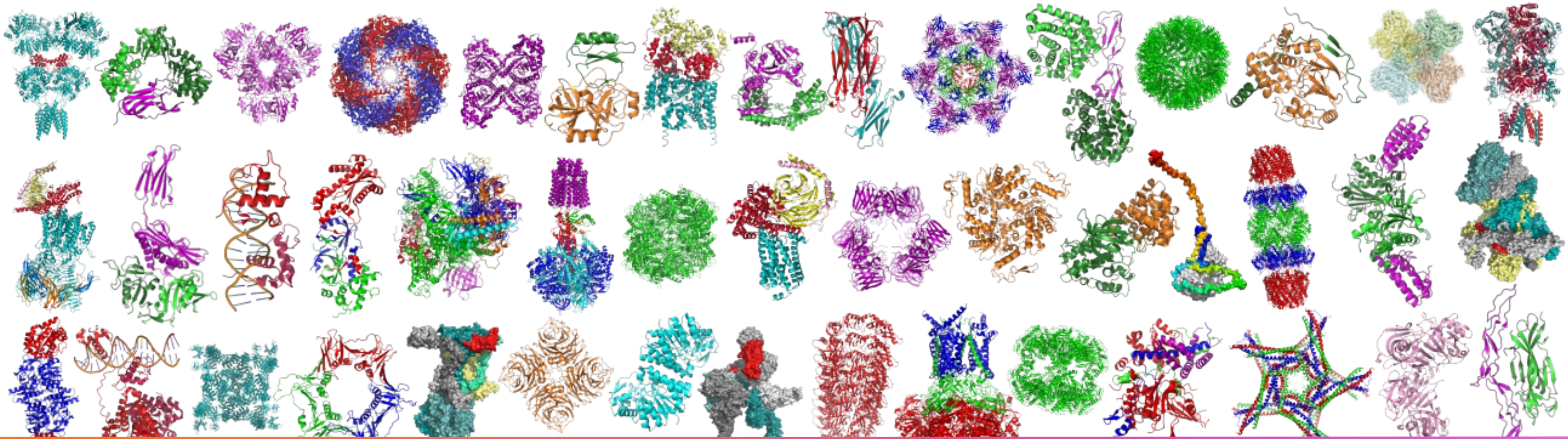
Limitations

- Captures only one conformation — unreliable for disordered, dynamic, or regulated regions.
- No direct information on ligands, cofactors, or alternative states.
- Confidence ≠ resolution — predicted models lack experimental validation.
- Unreliable for complexes.


Monomers (20.6%)
Complexes (79.4%)

Human protein complexes





How to choose a structure

- Prefer experimental structures when available; note the method (X-ray, cryo-EM, NMR) and resolution if given.
- Check what parts of the sequence are present and whether the construct includes truncations or mutations.
- Ensure the biological context matches your question: monomer or complex, active or inactive state, ligands present.
- Look at basic quality indicators provided in the PDB entry: missing residues, geometry outliers, clash scores.
- If no suitable experimental structure exists, use a high confidence predicted model and rely on per-residue confidence.
- There are ways to predict the structures of complexes, and they're getting a lot better, but they still get a lot wrong, so be careful.
- Pick the structure that best fits your task: detailed active site views need higher resolution; interaction or domain questions may not.
- Always confirm sequence alignment, chain identity and any bound molecules before using the structure.

ebi.ac.uk/pdbe/entry/pdb/7ala

New Chrome avail

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PDBe
Protein Data Bank in Europe

Home | Services | Documentation | Training

View PDBe entry by PDB entry ID

Explore

Examples: 2dn2 8b0x 7l35 5cxt 6qvt 3ah8

7ala

X-ray diffraction 1.846Å Released 13 Oct 2021 10.2210/pdb7ala/pdb

Download files | View files

Summary | Model Quality | Complexes | Macromolecules | Ligands and Environments | Domains | Text Annotations (AI) N/A | Citations

We have redesigned this page to enhance your experience. Take an on-page tour.

Entry title human GCH-GFRP inhibitory complex

Source organism *Homo sapiens*

Primary publication A hybrid approach reveals the allosteric regulation of GTP cyclohydrolase I.

Ebenhoch R, Prinz S, Kaltwasser S, Mills DJ, Meinecke R

Show all authors

Proc. Natl. Acad. Sci. U.S.A. 117 31838-31849 (2020)

PMID 33229582

DOI 10.1073/pnas.2013473117

Also associated with: 6z80, 6z85, 6z86, 6z87, 6z88

Show more associated entries

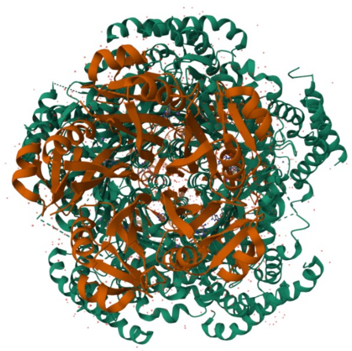
PDB model quality summary

Model geometry

Fit model/data

Percentile-sliders comparing the quality scores of a model with other models in the archive.

7ala



Preferred complex

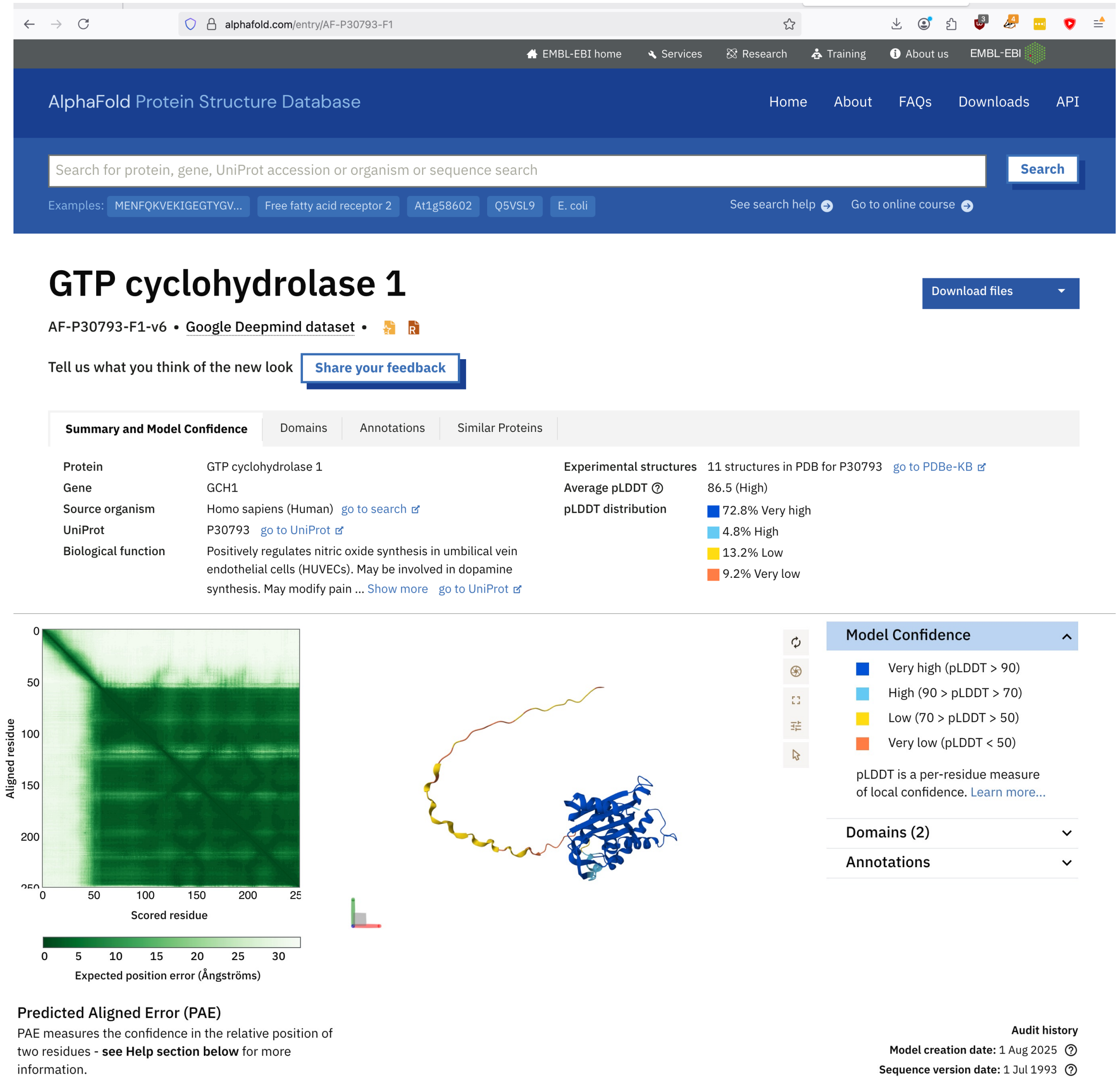
Macromolecules (2)

Ligands (3)

Domains (10)

Modifications (0)

alphafold.com

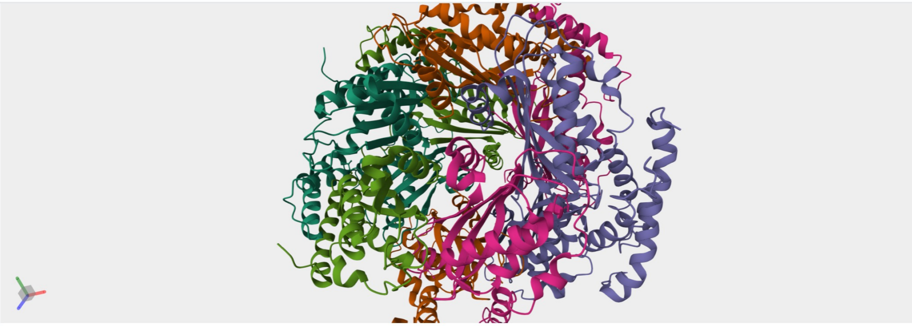


Pay attention to the model confidence!

uniprot.org


GCH1 - GTP cyclohydrolase 1
uniprot.org/uniprotkb/P30793/entry
UniProt
BLAST
Align
Peptide search
ID mapping
SPARQL
UniProtKB
Advanced
List
Search
Function
Names & Taxonomy
Subcellular Location
Disease & Variants
PTM/Processing
Expression
Interaction
Structure
Family & Domains
Sequence & Isoforms
Similar Proteins
P30793 · GCH1_HUMAN
Protein
GTP cyclohydrolase 1
Gene
GCH1
Status
UniProtKB reviewed (Swiss-Prot)
Organism
Homo sapiens (Human)
Amino acids
250 (go to sequence)
Protein existence
Evidence at protein level
Annotation score
5/5
Entry
Variant viewer 386
Feature viewer
Genomic coordinates
Publications
External links
History
Tools
Download
Add
Add a publication
Entry feedback
Function
Positively regulates nitric oxide synthesis in umbilical vein endothelial cells (HUVECs). May be involved in dopamine synthesis. May modify pain sensitivity and persistence. Isoform GCH-1 is the functional enzyme, the potential function of the enzymatically inactive isoforms remains unknown.
5 Publications

Structure



SOURCE -- Select --	IDENTIFIER	METHOD -- Select --	RESOLUTION	CHAIN	POSITIONS	LINKS	
PDB	1FB1	X-ray	3.10 Å	A/B/C/D/E	55-250	PDBe · RCSB-PDB · PDBj · PDBsum	· Folds
PDB	6Z80	EM	3.00 Å	A/B/C/D/E/F/G/H/I/J	41-250	PDBe · RCSB-PDB · PDBj · PDBsum	· Folds
PDB	6Z85	EM	2.90 Å	A/B/C/D/E/F/G/H/I/J	41-250	PDBe · RCSB-PDB · PDBj · PDBsum	· Folds
PDB	6Z86	X-ray	2.21 Å	A/B/C/D/E/F/G/H/I/J/K/L/M/N/O/P/Q/R/S/T	41-250	PDBe · RCSB-PDB · PDBj · PDBsum	· Folds
PDB	6Z87	X-ray	2.56 Å	A/B/C/D/E	41-250	PDBe · RCSB-PDB · PDBj · PDBsum	· Folds
PDB	6Z88	X-ray	2.69 Å	A/B/C/D/E/F/G/H/I/J	41-250	PDBe · RCSB-PDB · PDBj · PDBsum	· Folds
PDB	6Z89	X-ray	2.37 Å	A/B/C/D/E	41-250	PDBe · RCSB-PDB · PDBj · PDBsum	· Folds
PDB	7ALA	X-ray	1.85 Å	A/B/C/D/E	42-250	PDBe · RCSB-PDB · PDBj · PDBsum	· Folds
PDB	7ALB	X-ray	1.98 Å	A/B/C/D/E/F/G/H/I/J/K/L/M/N/O/P/Q/R/S/T	42-250	PDBe · RCSB-PDB · PDBj · PDBsum	· Folds
PDB	7ALC	X-ray	1.73 Å	A/B/C/D/E	42-250	PDBe · RCSB-PDB · PDBj · PDBsum	· Folds
PDB	7ALQ	X-ray	2.21 Å	A/B/C/D/E/F/G/H/I/J/K/L/M/N/O/P/Q/R/S/T	42-250	PDBe · RCSB-PDB · PDBj · PDBsum	· Folds
AlphaFold	AF-P30793-F1	Predicted			1-250	AlphaFold	· Folds

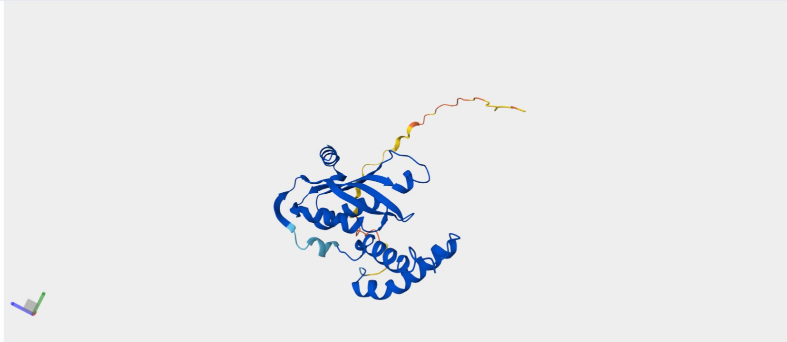
Structure

Select color scale
- Confidence
- Pathogenecity

Model Confidence:

- Very high (pLDDT > 90)
- Confident (90 > pLDDT > 70)
- Low (70 > pLDDT > 50)
- Very low (pLDDT < 50)

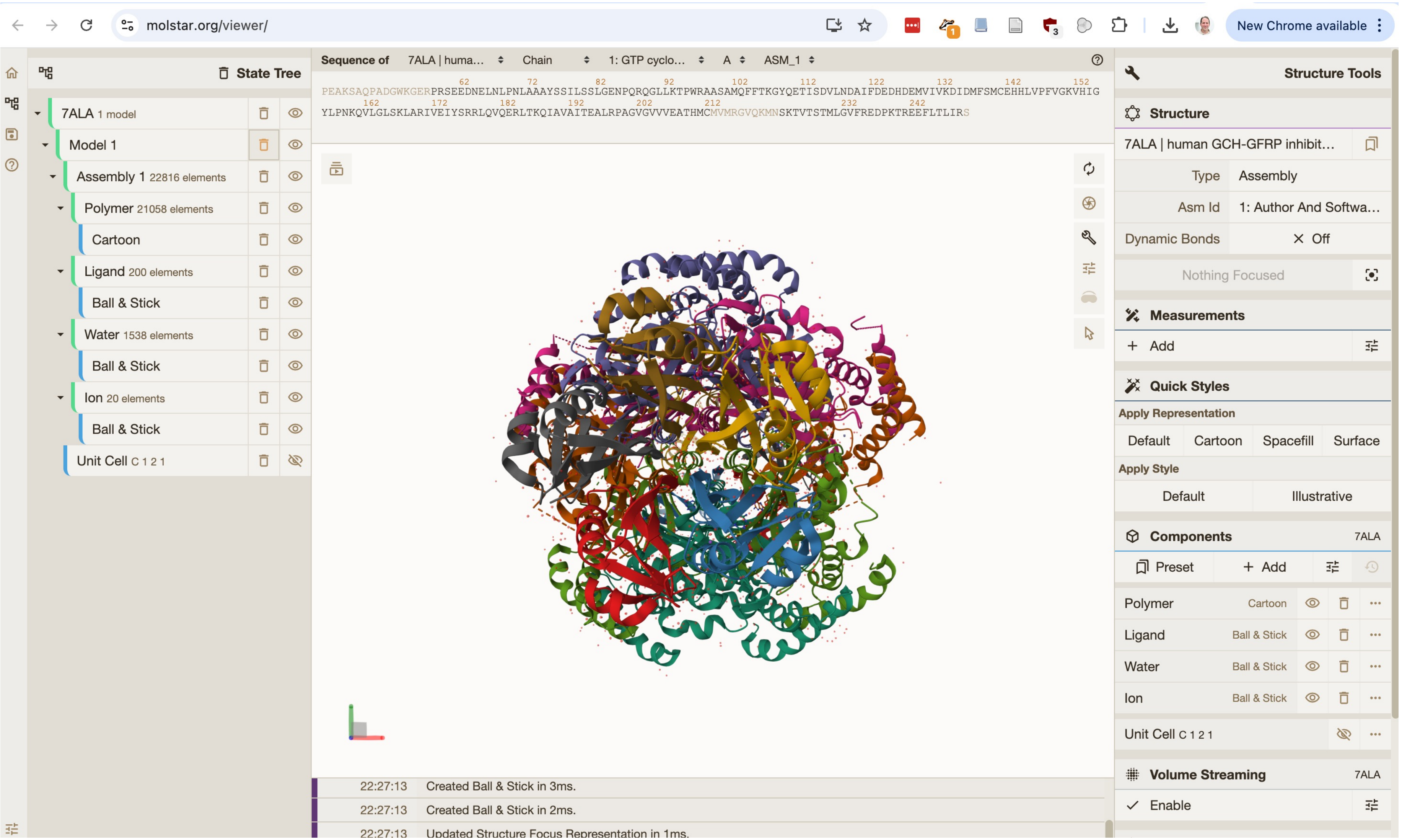
AlphaFold produces a per-residue confidence score (pLDDT) between 0 and 100. Some regions with low pLDDT may be unstructured in isolation.



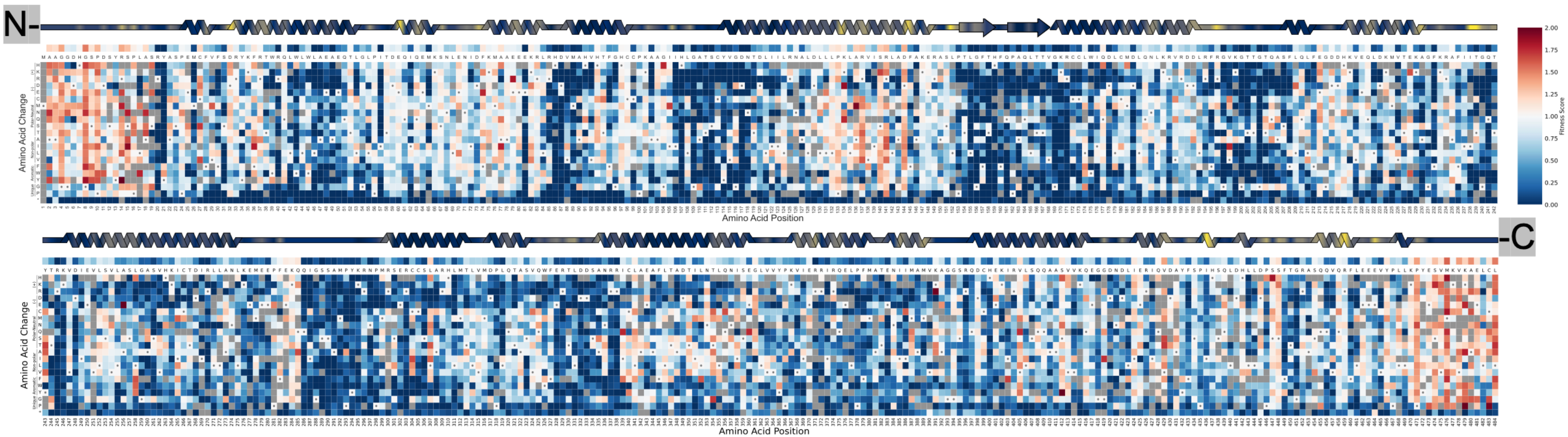
SOURCE -- Select --	IDENTIFIER	METHOD -- Select --	RESOLUTION	CHAIN	POSITIONS	LINKS	
PDB	1FB1	X-ray	3.10 Å	A/B/C/D/E	55-250	PDBe · RCSB-PDB · PDBj · PDBsum	· Foldseek
PDB	6Z80	EM	3.00 Å	A/B/C/D/E/F/G/H/I/J	41-250	PDBe · RCSB-PDB · PDBj · PDBsum	· Foldseek
PDB	6Z85	EM	2.90 Å	A/B/C/D/E/F/G/H/I/J	41-250	PDBe · RCSB-PDB · PDBj · PDBsum	· Foldseek
PDB	6Z86	X-ray	2.21 Å	A/B/C/D/E/F/G/H/I/J/K/L/M/N/O/P/Q/R/S/T	41-250	PDBe · RCSB-PDB · PDBj · PDBsum	· Foldseek
PDB	6Z87	X-ray	2.56 Å	A/B/C/D/E	41-250	PDBe · RCSB-PDB · PDBj · PDBsum	· Foldseek
PDB	6Z88	X-ray	2.69 Å	A/B/C/D/E/F/G/H/I/J	41-250	PDBe · RCSB-PDB · PDBj · PDBsum	· Foldseek
PDB	6Z89	X-ray	2.37 Å	A/B/C/D/E	41-250	PDBe · RCSB-PDB · PDBj · PDBsum	· Foldseek
PDB	7ALA	X-ray	1.85 Å	A/B/C/D/E	42-250	PDBe · RCSB-PDB · PDBj · PDBsum	· Foldseek
PDB	7ALB	X-ray	1.98 Å	A/B/C/D/E/F/G/H/I/J/K/L/M/N/O/P/Q/R/S/T	42-250	PDBe · RCSB-PDB · PDBj · PDBsum	· Foldseek
PDB	7ALC	X-ray	1.73 Å	A/B/C/D/E	42-250	PDBe · RCSB-PDB · PDBj · PDBsum	· Foldseek
PDB	7ALQ	X-ray	2.21 Å	A/B/C/D/E/F/G/H/I/J/K/L/M/N/O/P/Q/R/S/T	42-250	PDBe · RCSB-PDB · PDBj · PDBsum	· Foldseek
AlphaFold	AF-P30793-F1	Predicted			1-250	AlphaFold	· Foldseek

Structure visualisation tools

- Desktop applications (for advanced use):
  - **PyMOL** – publication-quality figures, scripting, extensive features
  - **ChimeraX** – powerful analysis, modern interface, integrates with EM and density data
- Browser-based tools (no installation):
  - **Mol*** – fast, accessible, ideal quick exploration
  - **iCn3D**, **NGL Viewer** – other capable web options


molstar.org/viewer/
New Chrome available
State Tree
7ALA 1 model
Model 1
Assembly 1 22816 elements
Polymer 21058 elements
Cartoon
Ligand 200 elements
Ball & Stick
Water 1538 elements
Ball & Stick
Ion 20 elements
Ball & Stick
Unit Cell C 1 2 1
Sequence of
7ALA | huma...
Chain
1: GTP cyclo...
A
ASM_1
PEAKSAQPADGWKGERPRSEEDNELNLPNLAAAYSSILSSLGENPQRQGLLKTPWRAASAMQFFTKGYQETISDVLNDAIFDEDHDEMVIVKDIDMFSMCEHHLVPFVGKVHIG
YLPNKQVLGLSKLARIVEIYSRRLQVQERLTKQIAVAITEALRPAGVGVVVEATHMCMVMRGVQKMNSKTVTSTMLGVFREDPKTREEFLTLIRS
22:27:13 Created Ball & Stick in 3ms.
22:27:13 Created Ball & Stick in 2ms.
22:27:13 Updated Structure Focus Representation in 1ms.
Structure Tools
Structure
7ALA | human GCH-GFRP inhibit...
Type
Assembly
Asm Id
1: Author And Softwa...
Dynamic Bonds
Off
Nothing Focused
Measurements
Add
Quick Styles
Apply Representation
Default
Cartoon
Spacefill
Surface
Apply Style
Default
Illustrative
Components
7ALA
Preset
Add
Polymer
Cartoon
Ligand
Ball & Stick
Water
Ball & Stick
Ion
Ball & Stick
Unit Cell C 1 2 1
Volume Streaming
7ALA
Enable

ADSL heatmap



New Results

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Mechanistic Modelling of Recessive Disease through Allelic Integration of Variant Effects

Posted August 16, 2025.

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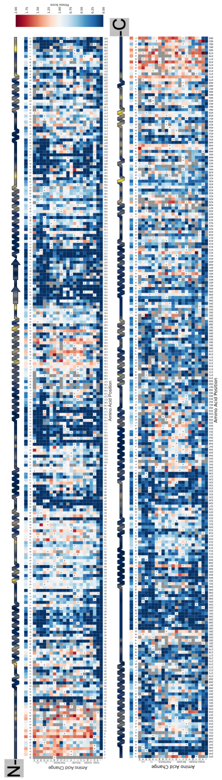
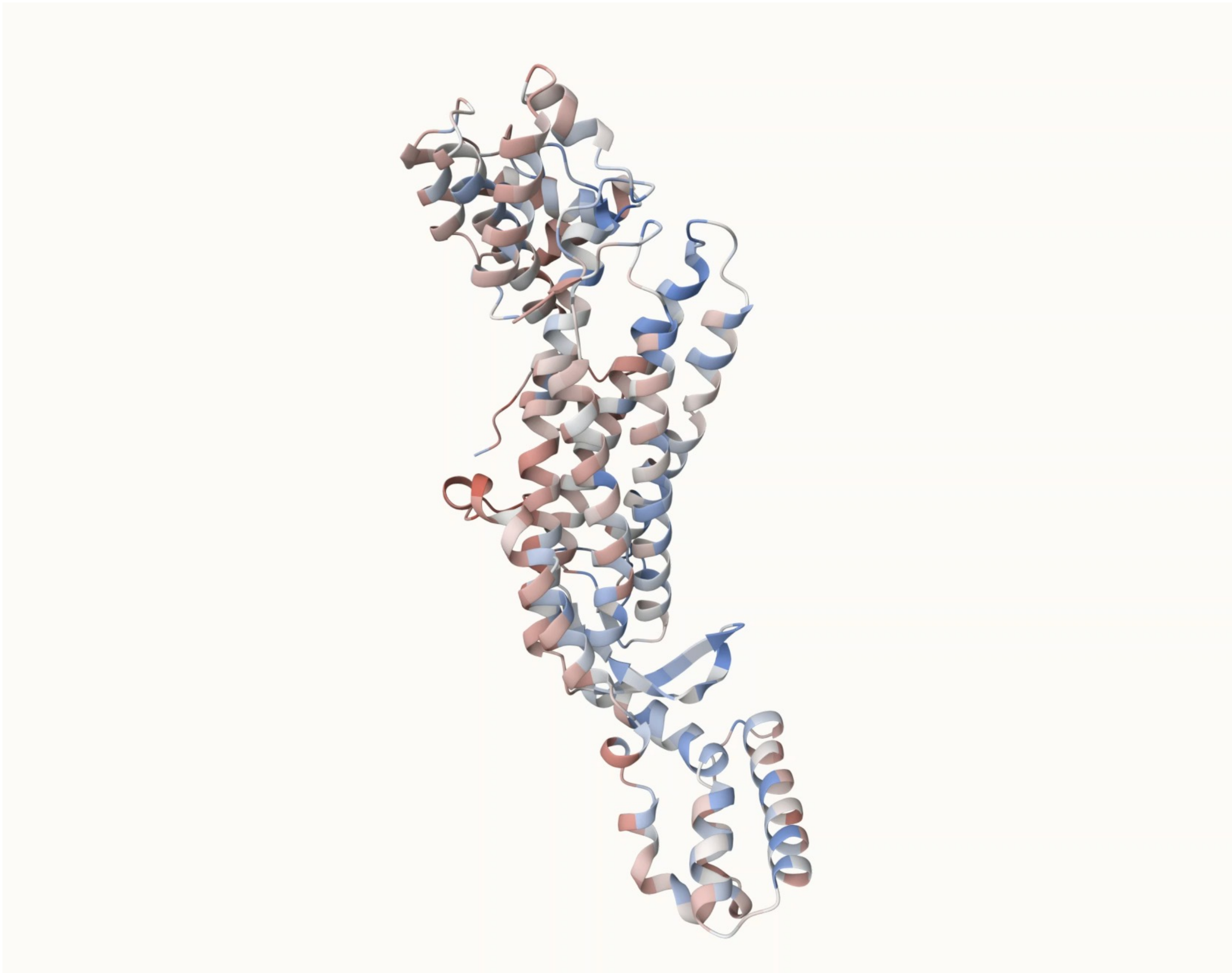
Hasan Çubuk, Marcin Plech, Vahid Aslanzadeh, Marie Zikanova, Vaclava Skopova, Stanislav Kmoch, Yuxin Shen, Joseph A. Marsh, Grzegorz Kudla

doi: https://doi.org/10.1101/2025.08.15.670494

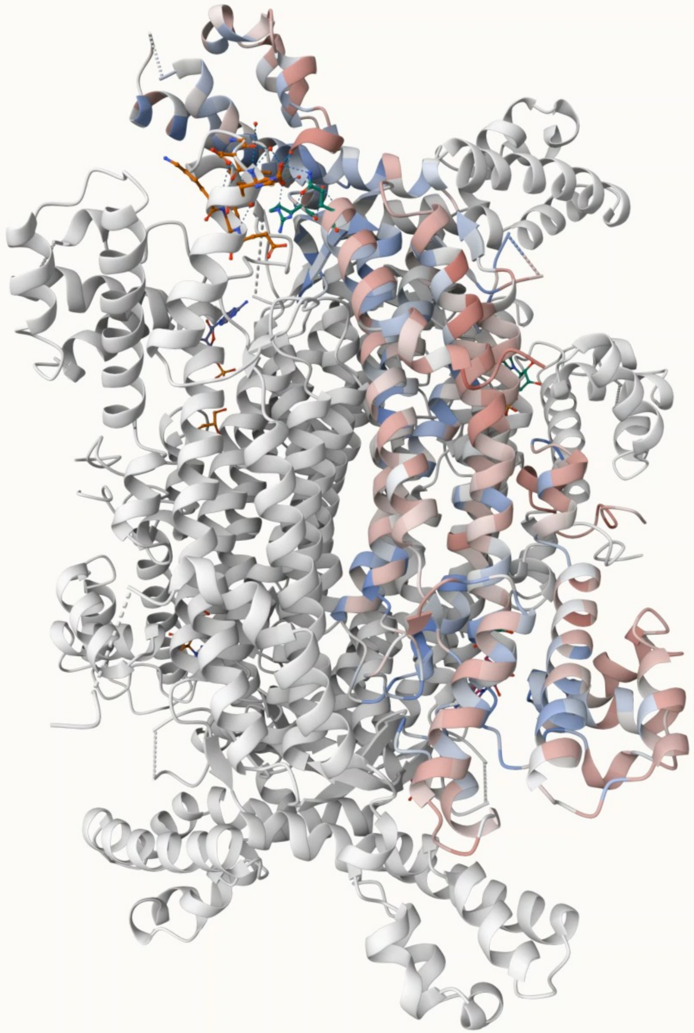
This article is a preprint and has not been certified by peer review [what does this mean?].



AlphaFold model

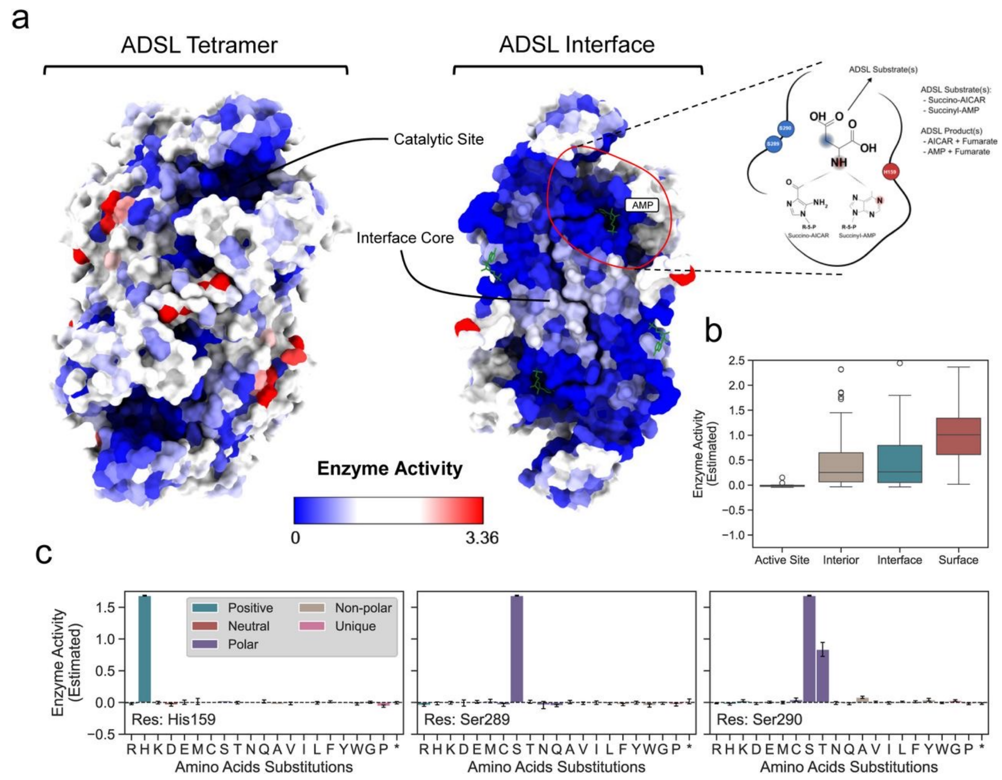


Crystal structure



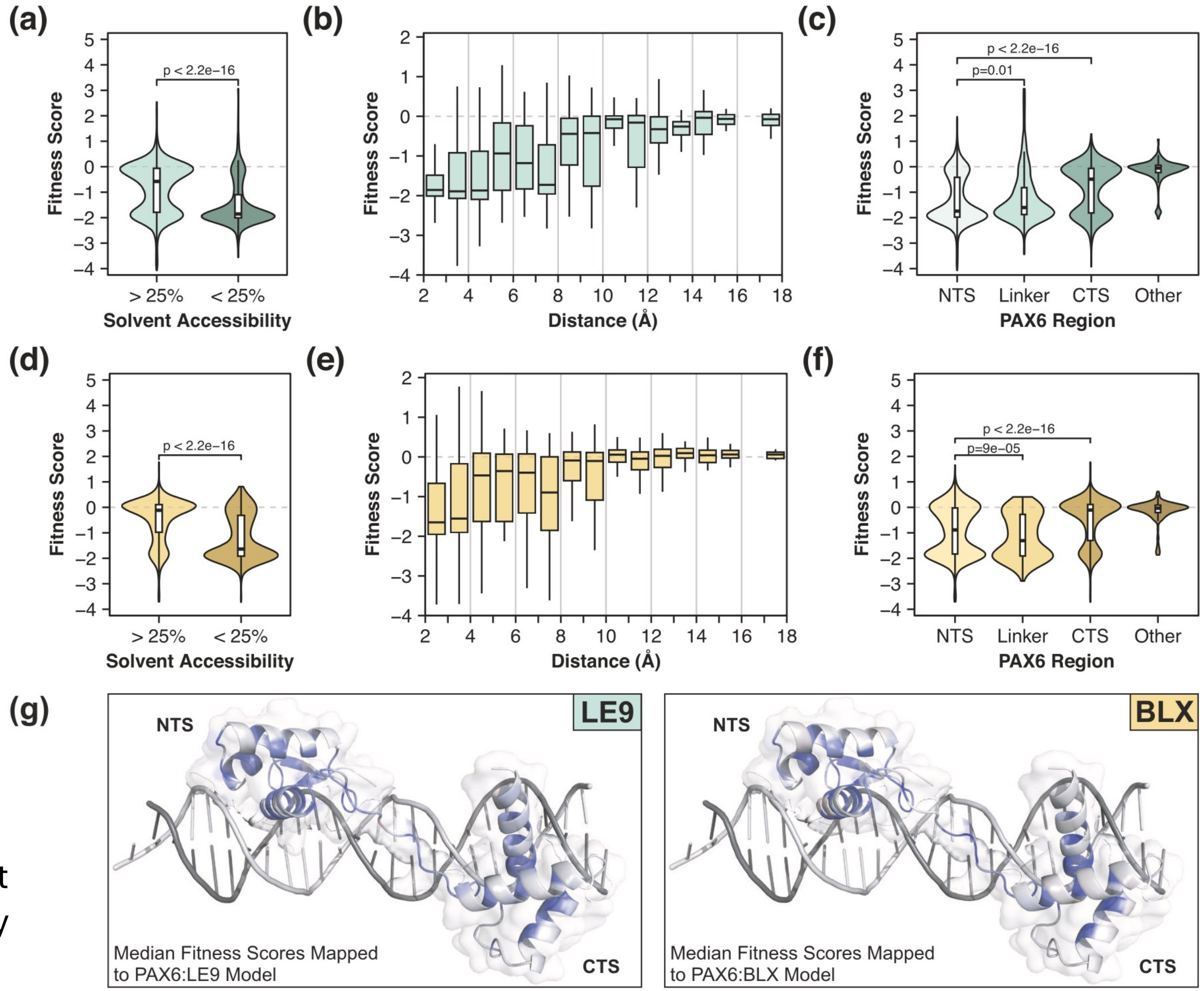
ADSL structural context

- Mapping MAVE scores (enzyme activity) onto the ADSL tetramer highlights a simple structural pattern:
- **Active site residues** are almost completely intolerant to substitution, consistent with their direct catalytic roles.
- **Buried core and interface residues** show substantial constraint, reflecting the need to maintain tetramer assembly and overall fold.
- **Surface residues** are far more permissive, supporting substitutions without major functional consequences.
- The catalytic residues **His159**, **Ser289**, and **Ser290** illustrate this sharply: nearly all substitutions eliminate activity, except Ser290Thr which retains partial function.



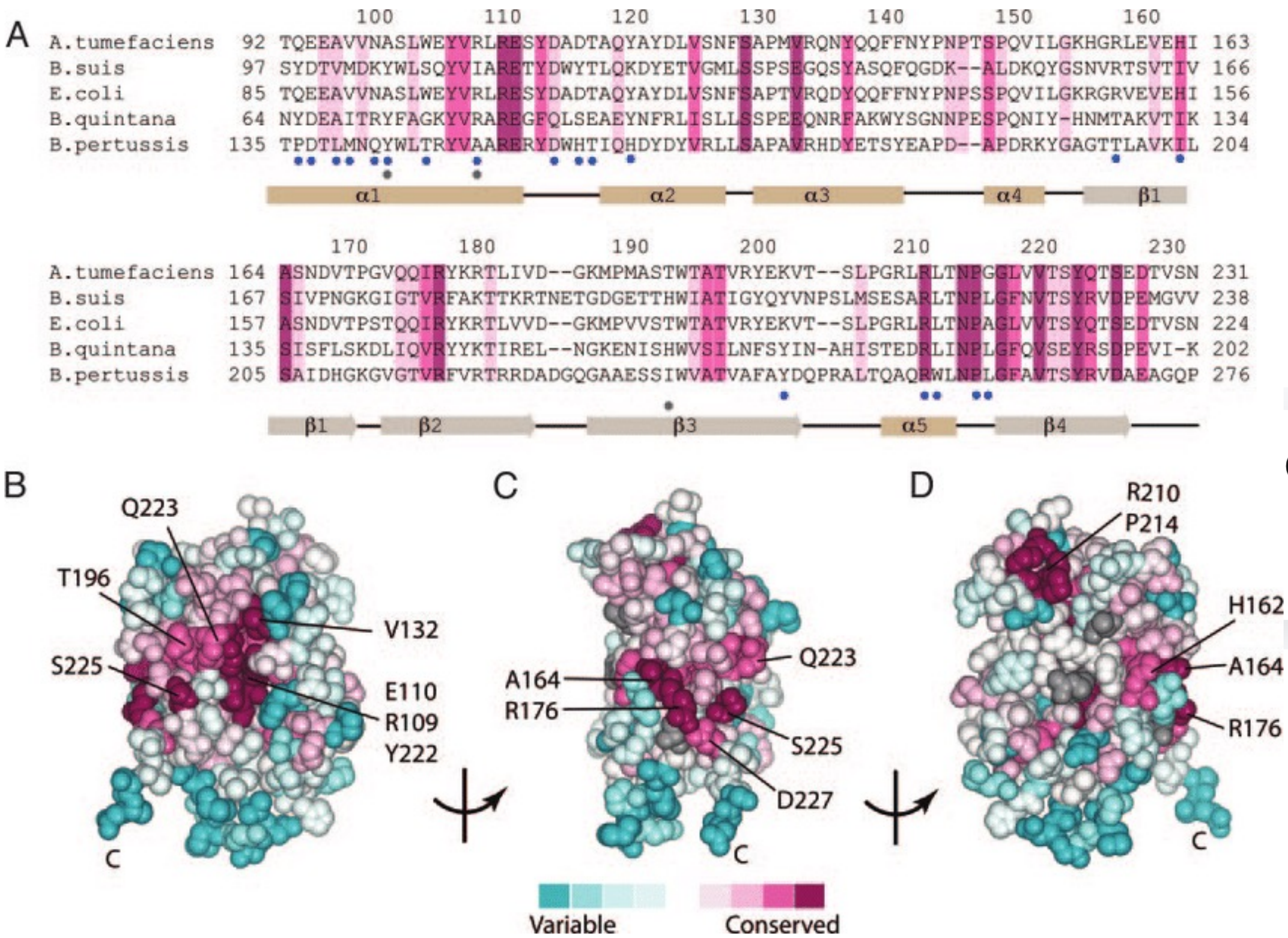
PAX6 structural context

- Mutations **closest to the DNA interface** show the strongest loss of binding
- **Buried residues in the domain cores** are strongly constrained, consistent with their contribution to folding stability.
- **Surface-exposed residues** tolerate substitution much better, giving the pale "neutral" banding across the map.
- The two subdomains behave differently: the **N-terminal subdomain** is globally more sensitive to mutation than the C-terminal, in line with known functional asymmetry.
- Positions with striking gain or inversion of effect between the two DNA targets (e.g. at **positions 50, 54, 71**) cluster at structurally DNA-proximal sites, consistent with sequence-specific changes in contact geometry.



McDonnell et al (2024) Mol Syst Biol

Evolutionary conservation



ConSurf Temporarily Unavailable

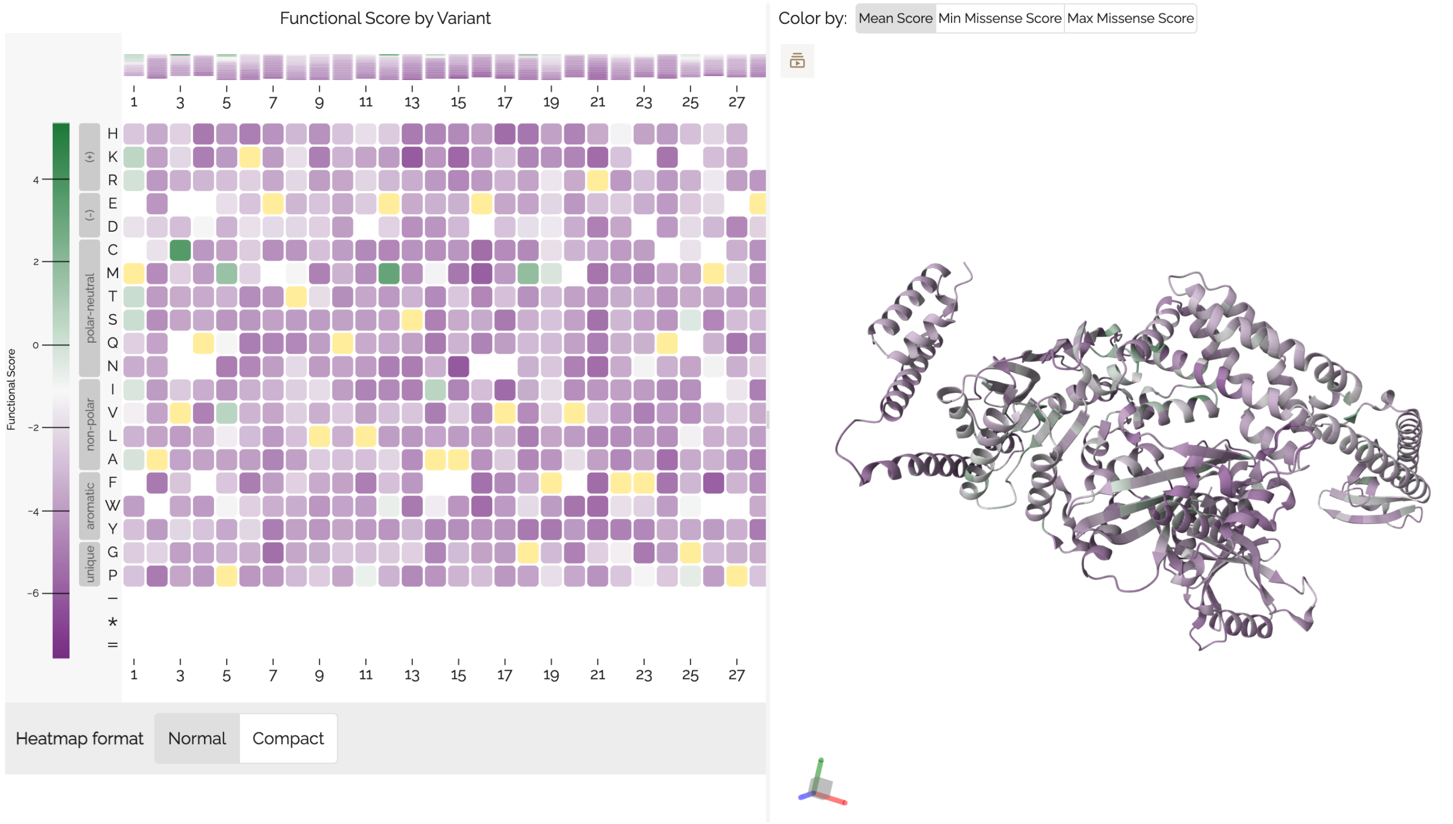
ConSurf is down due to technical issues.

You can try to run ConSurf on Google Colab.

Visualising variant effect scores

MaveDB built in visualisation

MSH2 LOF scores (HAP1)



Try it yourself

https://sites.google.com/view/mave2025-lectures

1. Download a MAVE dataset
   - Choose your own at MaveDB, or pick a prepared one from the link above
   - Make sure to pick something based on single amino acid substitutions (most are)
2. Select a corresponding structure to visualize
   - Select one from Uniprot (or download from the link)
   - Warning – sometimes there are mismatches in residue numbering between the MAVE and structure
3. Combine the variant scores and PDB in the Colab link
   - Writes the residue-averaged scores into the PDB file making visualization easier
   - Need a Google account
4. Load the structure in *mol** and visualize MAVE scores
5. **What is your mechanistic interpretation?**